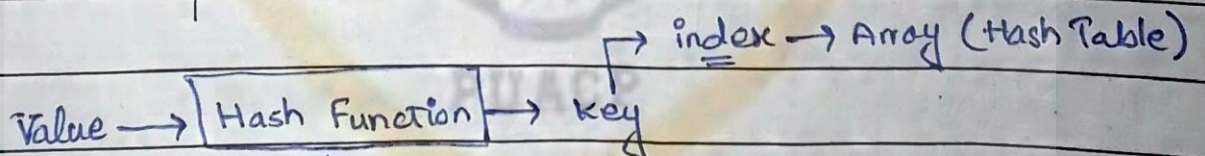


Hashing

"A technique in which addition of value, deletion, searching, and a random is done in constant time or $O(1)$."

Array (unsorted)	Array (sorted)	LinkedList	Tree	AVL
Add: $O(1), O(n)$	Add: $O(n)$	Add: $O(n)$	Add: $O(n)$	Add: $O(\log_2 n)$
Access: $O(1)$	Access: $O(1)$	Access: $O(n)$	Access: $O(n)$	Access: $O(\log_2 n)$
Search: $O(n)$	Search: $O(\log_2 n)$	Search: $O(n)$	Search: $O(n)$	Search: $O(\log_2 n)$
Delete: $O(n)$	Delete: $O(n)$	Delete: $O(n)$	Delete: $O(n)$	Delete: $O(\log_2 n)$

Hashing
Add: $O(1)$
Access: $O(1)$
Search: $O(1)$
Delete: $O(1)$



$\rightarrow$ uses some techniques to convert value to key.

$\rightarrow$ if key is unique for each value, then it's perfect hashing function.

Many Hashing Methods are used in Hash Function.

Hash Functions:

i) Division (Also called Modulus)

- uses mod to find key
value mod n

$$= \text{value} \% n$$

→ n maybe equal to Hash table size.

→ it can be a specific value.

e.g.

suppose, Hash Table size is 5.

values: 10, 7, 3, 11, 4 , Hash Function: value mod 5

Hash Table:

0	10
1	7
2	3
3	11
4	4

① $10 \% 5 = 0$

② $7 \% 5 = 2$

③ $3 \% 5 = 3$

④ $11 \% 5 = 1$

⑤ $4 \% 5 = 4$

ii) Folding:

i) shift Folding: suppose Hash Table Size = 1,000.

value = 123456789

$$\text{key} = 123 + 456 + 789$$

$$= 1368$$

$$\downarrow 1368 \% 1000$$

$$= 368 \rightarrow \text{index.}$$

i) Boundary Folding:

suppose, Hash Table Size = 1,000

value = 123456789

$$\text{key} = 123 + 456 + 789$$

$$= 1566$$

$$= 1566 \% 1000$$

$$= 566 \rightarrow \text{index.}$$

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Collision Resolution

i) Open Addressing (Probing)

a) Linear Probing $h(k)+1$, on each collision.

suppose, Hash Table size = 5

values = 10, 6, 5, 4, 2

HF = $X \bmod 5$

	0	10
	1	6
collision Resolve $\rightarrow$	2	5
collision Resolve. $\rightarrow$	3	2
	4	4

$$10 \% 5 = 0$$

$$6 \% 5 = 1$$

(collision) $5 \% 5 = 0$

$$4 \% 5 = 4$$

(collision) $2 \% 5 = 2$

$\rightarrow$ due to linear Probing causes ~~linear~~ Primary clustering.

b) Quadratic Probing

$h(k) + n^2$, on each collision.

$n = 1, 2, 3, \dots$

suppose, Hash Table size = 5

value = 10, 6, 5, 4, 2

HF = $X \bmod 5$

0	10
1	6
2	
3	
4	5

$$= 0 + (1)^2$$

$$= 1$$

$$= 1 + (2)^2$$

$$= 1 + 4 = 5$$

$$= 5$$

$$= 5 \% 5 = 0$$

$$10 \% 5 = 0$$

$$6 \% 5 = 1$$

$$5 \% 5 = 0$$

$$4 \% 5 = 4$$

Secondary clustering

$$= 0 + (3)^2$$

$$= 9 \% 5 = 4$$

$\leftarrow$ once again collision due to resolution.



			3%. S = 3
		4	IST. S = 0
			17%. S = 2
4	3		

c) double hashing

- if collision occurs for $H(k)$.
- $H(k) + H_2(k) = \text{new key}$.
↳ 2nd Hashing Function.
- still many collisions and clustering will occur.

ii) Chaining (Separate Chaining)

suppose, Hash Table size = 5

values = 10, 5, 3, 15, 17, 2, 18

$$H(k) = X \bmod 5$$

0	→	10	→	5	→	15	1	$10 \% 5 = 0$
1	/							$5 \% 5 = 0$
2	→	17	→	2	1			$3 \% 5 = 3$
3	→	3	→	18	1			$15 \% 5 = 0$
4	/							$17 \% 5 = 2$
								$2 \% 5 = 2$
								$18 \% 5 = 3$

iii) Coalesced Chaining

suppose, Hash Table size = 5

value = 10, 5, 3, 15, 17

$$H(k) = X \bmod 5$$

$$10 \% 5 = 0$$

$$5 \% 5 = 0$$